

Contents

1	Introduction	1
1.1	Research Question	2
1.2	Data	2
2	Time Series Analysis	4
2.1	Decomposition	4
2.2	Autocorrelation	5
2.3	Holt–Winters Forecasting	6
2.4	Random Walk	8
2.5	Augmented Dickey–Fuller Test	9
2.6	SARIMA Forecasting Models	10
2.7	Models Forecast Accuracy 2026	10
2.8	GARCH Models	12
3	Limitations & Conclusion	14

List of Tables

1	Descriptive statistics: German weekly wholesale electricity prices (€/MWh)	3
2	Estimated Holt–Winters smoothing parameters	7
3	Augmented Dickey–Fuller test results	10
4	Out-of-sample forecast accuracy for the 2026 evaluation period	12
5	95% confidence intervals for the GARCH(1,1) parameters	13

List of Figures

1	German Energy Wholesale Prices: October 2018 – January 2026	3
2	Comparison level series and log-transformed series	4
3	Weekly price series decomposition	5
4	Correlogram and Partial Correlogram of the random component	6
5	2-year forecast using Holt–Winters method	7
6	Holt–Winters residuals ACF plot	8
7	ACF of the first-differenced log series	9
8	2026 price forecast for <i>full period model</i> , <i>post-crisis model</i> , and Holt–Winters	11
9	Correlogram of <i>full period</i> and <i>post-crisis</i> SARIMA squared residuals . . .	13

List of Abbreviations

ACF	Autocorrelation Function
PACF	Partial Autocorrelation Function
HW	Holt–Winters
ARIMA	Autoregressive Integrated Moving Average
SARIMA	Seasonal Autoregressive Integrated Moving Average
GARCH	Generalized Autoregressive Conditional Heteroskedasticity
MAD	Mean Average Deviation
RMSE	Root Mean Squared Error

1 Introduction

The development of energy prices has at times attracted massive public attention. The globalization of energy markets and supplies have contributed to large interdependencies. Since domestic energy prices are largely determined by global commodity prices, energy producing countries can leverage and influence global price structures, steering market prices upwards. This phenomenon is not new for German consumers. Large parts of the population still vividly recall the oil price crisis of 1973 and 1979, which saw global oil prices multiply in a matter of weeks after OPEC production cuts. The 24th of February, the day that Russia started the invasion and war on Ukraine, marked the starting point of yet another energy price crisis that sent shock waves through the European energy markets (Eddy & Aldermann, 2022).

Such price shocks are at odds with the imperative of economic stability. Energy prices directly affect inflation data that consumers, investors, and central banks alike rely on. They influence how a variety of future investments are evaluated, including energy-intensive factories and facilities. The Russian attack and subsequent price increases serve as a vivid example of energy price shocks. When the European Union jointly sanctioned Russian energy imports, the effects were felt across the continent. Within a matter of days, energy prices exploded. Heating, production and manufacturing costs rose sharply. Media outlets were quick to report a new European energy crisis. Governments imposed costly countermeasures to limit the fallout of rising and persistently high energy prices.

The remainder of the paper proceeds as follows. In Section [2](#), I apply time series analysis and forecasting methods. Section [2.6](#) details the application of SARIMA forecasts with different training windows. Lastly, I conclude in section [3](#) discussing potential limitations and results of the analysis.

1.1 Research Question

When the high electricity prices eventually normalized, economic concerns remained. Amongst them was the question what such crisis would mean for the thousands of companies, individuals and policymakers in Germany involved in energy price forecasting for investment and trading as well as policy and employment decisions. Therefore, the paper investigates the following question:

Does including the 2022 energy crisis period degrade German electricity price forecast accuracy?

This inquiry promises to be especially fruitful, given that recent developments in the Middle East have substantially driven energy prices upwards again. If results suggest that including crisis periods degrade forecasting accuracy, energy price forecasting may ideally rely on pre- or post-crisis training data. The opposite result is equally informative. If including crisis periods does not degrade accuracy, energy price forecasting can make use of the vast amount of historical price data available.

1.2 Data

The data used in this project comes from the publicly available SMARD electricity and gas market dataset. It is provided directly from Federal Bureau for Grids and Energy Systems. The data covers the period from October 2018, following a bidding-zone split, until the first week of August 2026. The dataset used for this analysis has weekly frequency, which amounts to 410 observations throughout the period with no missing observations. The main value is the wholesale price in € per Megawatt hour (MWh) of electricity in the Germany and Luxembourg price region. For reference; a two person household consumes roughly 3.5 MWh per year of electricity (Statistisches Bundesamt, 2023).

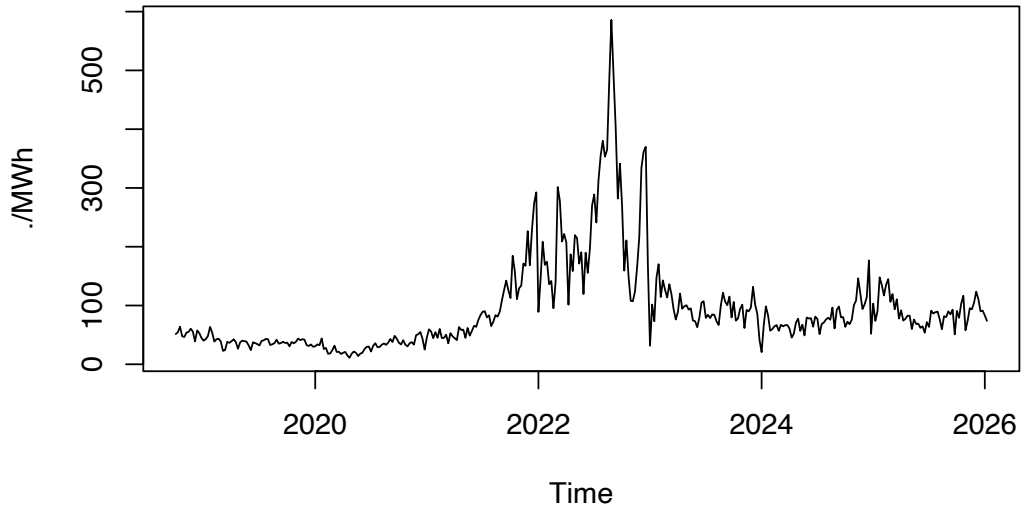


Figure 1: German Energy Wholesale Prices: October 2018 – January 2026

What becomes visible in the Figure 1 are the previously discussed effects of the Russian invasion of Ukraine in February of 2022. The graph clearly shows the resulting rise in German energy prices associated with the invasion. Energy prices stabilized afterwards, though priced higher compared to pre-crisis levels.

Minimum	1st quartile	Median	Mean	3rd quartile	Maximum
11.19	42.77	73.85	93.23	106.50	585.92

Table 1: Descriptive statistics: German weekly wholesale electricity prices (€/MWh)

The descriptive statistics in Table 1 show that prices fluctuate heavily throughout the observed series. The lowest price is recorded at 11.19 €/MWh, while the highest price sits around 585.92 €/MWh, about 52 times higher. Such fluctuation in the relevant variable suggests that log-transformation of the series is reasonable. The comparison between the series is visualized in Figure 2.

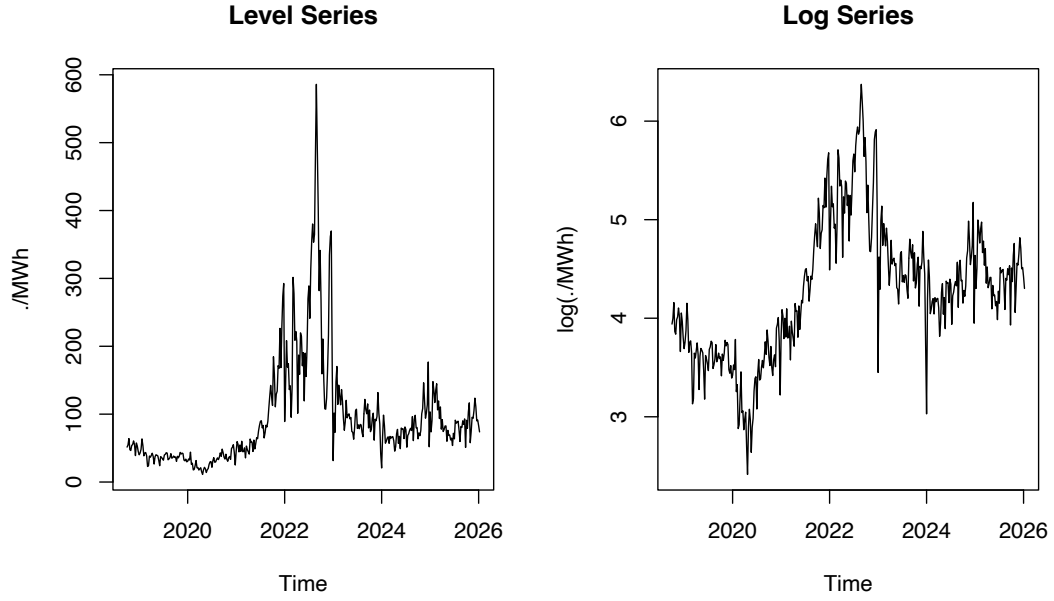


Figure 2: Comparison level series and log-transformed series

2 Time Series Analysis

The following time series analysis adopts methods from Paul Cowpertwait and Andrew Metcalfe (2009). The analysis will use the log-transformed series. To answer the question at hand, the analysis will focus especially on different forecasting methods and their suitability for the given series. Aside from chapter 2.6 and 2.7, the following steps are applied to only 379 weekly observations from October 2018 until the first week of January in 2026. Observations from 2026 will later be used as a baseline to compare forecasting accuracy of two models with different training windows.

2.1 Decomposition

Applying additive decomposition identifies and separates trend, seasonal and random component within the given series.

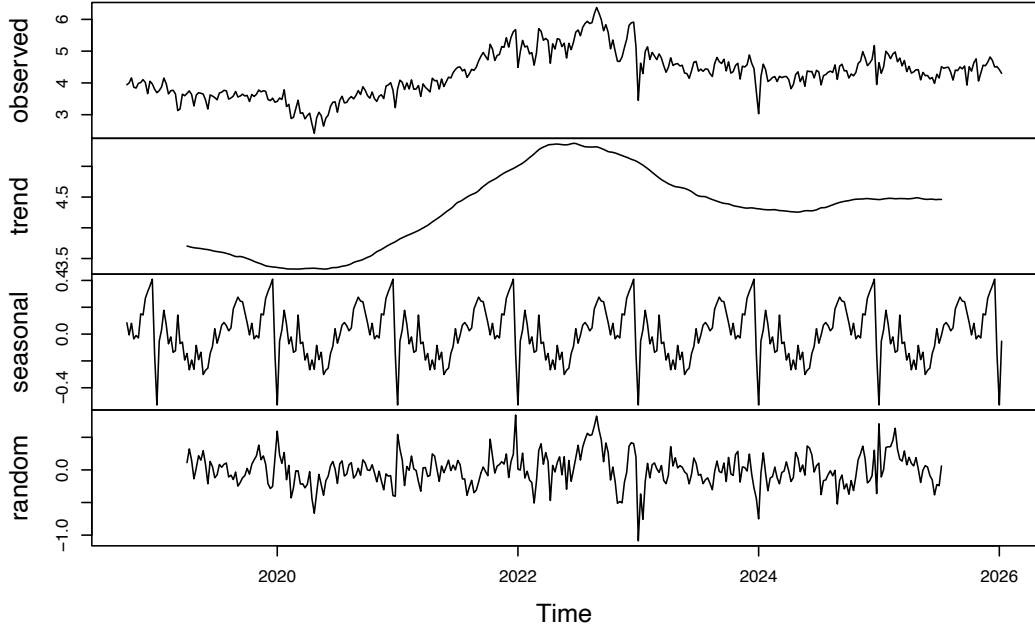


Figure 3: Weekly price series decomposition

As expected, the trend component exhibits an upward slope around the time of the energy crisis and a decline in subsequent months. The seasonal component shows a recurring annual pattern, potentially described by seasonal fluctuations in demand or varying renewable energy production. Fluctuation and spikes in the random component motivate further analysis of the autocorrelation.

2.2 Autocorrelation

The typical autocorrelation function (ACF) estimates the linear correlation between the random component at time t and the same component k periods earlier. Similarly, the PACF plot shows the correlation at lag k after removing the correlations stemming from shorter lags. The following ACF plot in Figure 3 therefore allows us to evaluate whether autocorrelation in the random component remains after removing the trend and seasonality.

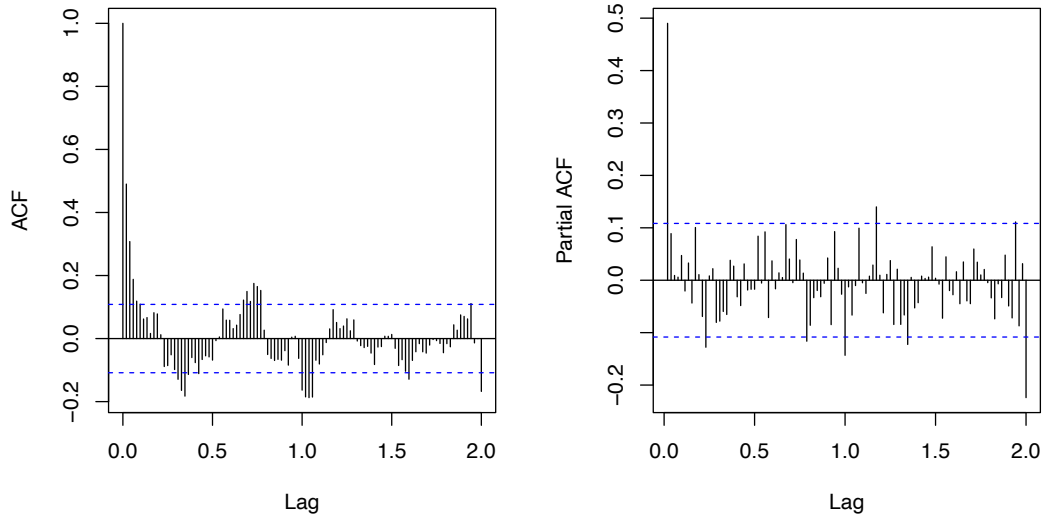


Figure 4: Correlogram and Partial Correlogram of the random component

The ACF and PACF structure suggests that the random component autocorrelation does not resemble white noise. ACF and partial ACF (PACF) show significant spikes exceeding the 95 % confidence bands at different lags. The PACF shows a spike at lag 1 with rapidly declining lag afterwards. This observation is consistent with what to expect for an AR(1) component. Importantly, the ACF gradually declines while showing annually recurring spikes. The observable annual seasonal autocorrelation motivates a *Seasonal* Autoregressive Integrated Moving Average (SARIMA) specification later in the analysis.

2.3 Holt–Winters Forecasting

I first introduce a Holt–Winters (HW) forecasting approach as a baseline forecasting model before moving to other forecasting approaches that can be compared against this baseline forecast. The HW forecasting method allows the seasonal variation and trend, described in terms of a level and slope, to change over time and estimates these features by exponentially weighted averages (Cowpertwait & Metcalfe 2009, P. 67).

Parameter	Estimate
$\hat{\alpha}$	0.3914
$\hat{\beta}$	0.0028
$\hat{\gamma}$	0.4575

Table 2: Estimated Holt–Winters smoothing parameters

As shown in Table 2, the optimal values of the smoothing parameters, obtained by minimizing the one-step-ahead prediction errors, are approximately $\hat{\alpha} = 0.392$, $\hat{\beta} = 0.003$, and $\hat{\gamma} = 0.457$. This suggests that the level and seasonal variation adapt rapidly whereas the trend is slow to do so (Cowpertwait & Metcalfe, 2009). Using these values, we can construct a 2-year forecast using HW, shown in Figure 5. Due to the construction of the HW approach, the difference between the upper and lower intervals becomes greater for longer time horizons.

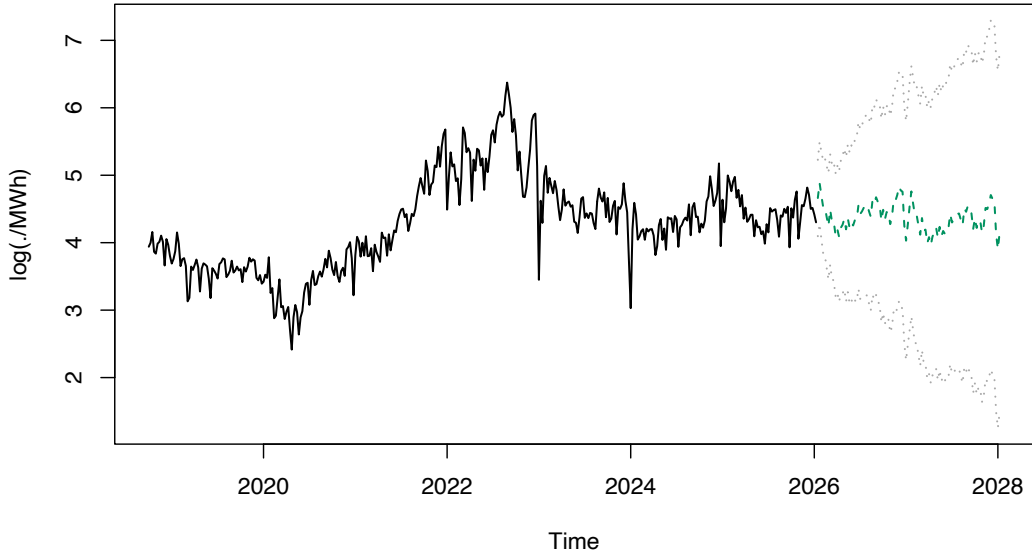


Figure 5: 2-year forecast using Holt–Winters method

What remains to be investigated is whether the HW forecasting approach is effective in minimizing the residual errors. Ideally, if the model accounts for all serial correlation in the model, the ACF plot should show a white noise structure with no obvious patterns.

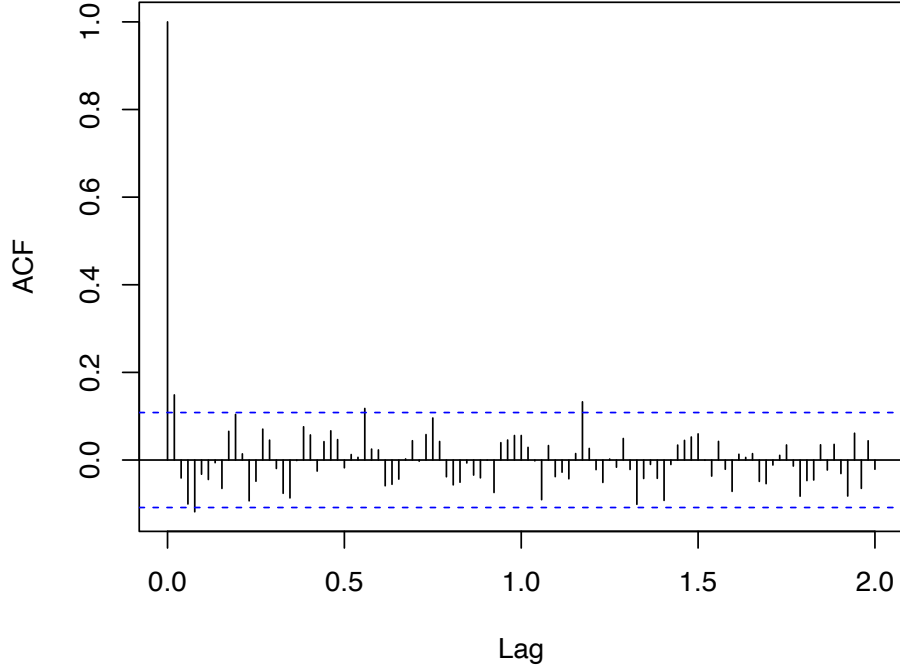


Figure 6: Holt–Winters residuals ACF plot

Figure 6 displays the ACF plot for the HW residuals. They show that autocorrelation has decreased compared to the naïve decomposition approach but nevertheless does not completely remove autocorrelation. The residuals are still not completely approximated by white noise, which motivates fitting the following models.

2.4 Random Walk

What remains to be tested is evidence for a $MA(q)$ component. This can be established using the ACF of the first-differenced log series. The first-order differences of a random walk are a white noise series (Cowpertwait & Metcalfe 2009, p. 74). Therefore, the correlogram should show white noise structure given that the series can be modelled by a random walk. Formally:

$$x_t = x_{t-1} + w_t \quad (1)$$

$$\nabla x_t = x_t - x_{t-1} = w_t \quad (2)$$

where x_t is the value of the observed series at time t , x_{t-1} its value one period earlier, $\nabla x_t = x_t - x_{t-1}$ the first difference, and w_t a white-noise process with $E(w_t) = 0$, constant variance $\text{Var}(w_t) = \sigma^2$, and no autocorrelation across time.

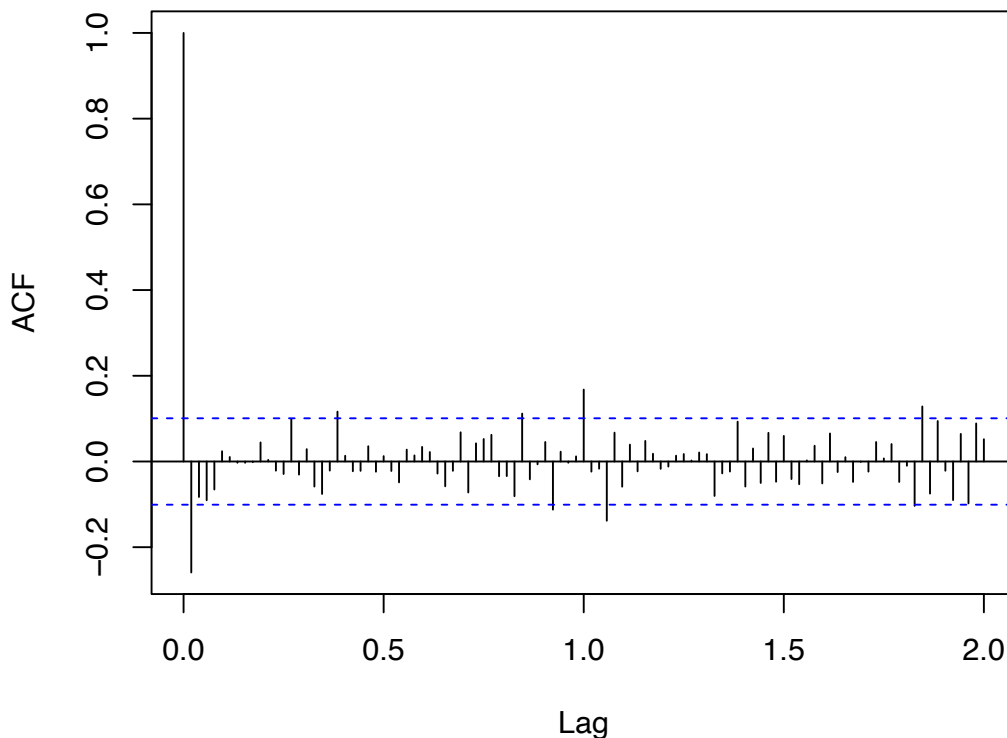


Figure 7: ACF of the first-differenced log series

The ACF plot of the first-differenced series in Figure 7 is closer to white noise but still exhibits autocorrelation at lag 1, which becomes substantially smaller and insignificant at subsequent lags. This provides sufficient reason to consider the MA(1) component

2.5 Augmented Dickey–Fuller Test

We have so far found evidence for seasonal annual autocorrelation, AR(1) through the PACF of the residuals of the decomposed series and MA(1) through the ACF plot of the first-differenced series. What remains to be established is whether the series is integrated. The Dickey-Fuller Test in R can help determine the potential order of integration, d . This requires that the log-level series contains a unit root and the first difference is stationary.

Series	ADF statistic	Lag order	p -value
Log-price series, y_t	-1.6908	7	0.7076
First-differenced series, Δy_t	-9.8294	7	< 0.01

Table 3: Augmented Dickey–Fuller test results

Table 3 shows the results of the Dickey-Fuller Test for first-differenced series. The result suggests that H_0 can be rejected ($p < 0.01$), hence the differenced series is stationary.^[1]

2.6 SARIMA Forecasting Models

Findings from previous chapter suggest a SARIMA model for forecasting. The following section will find and apply SARIMA(p,d,q)(P,D,Q) using the `get.best.arma` function to confirm or reject the previous findings. I thereby control whether the proposed SARIMA structure with AR(1) and MA(1) holds. I test this with `get.best.arma` and maximum order of (2,2,2,2,2,2) to account for numerous possible combinations.^[2] This returns the result that the optimal specification is SARIMA (1,1,1) (0,0,1). The next steps forecast the SARIMA model based on the full period and post–crisis period model and compare their out–of–sample performance for 2026 forecasts. This allows to test the two models forecasts against actual energy price developments in 2026.

2.7 Models Forecast Accuracy 2026

The first approach applies the SARIMA(1,1,1)(0,0,1) specification to the full period, using all observed values from October 2018 until the end of December 2025. As such, the full period includes the 2022 price shock and regime change.

The second approach applies the same SARIMA specification to a post-crisis training window. The selection criterion for the start of the post–crisis model is not disambiguous. I chose the post-crisis window to start in Juli 2023 and run until the end of December 2025.^[3]

¹Hypothesis comes from test output

²Given the previous findings, maximum order function of (1,1,1,1,1,1) would likely be sufficient

³One could choose a later starting point. This would however sacrifice observations which are already fewer in post–crisis window. This is the main motivation to use an early starting date

To answer the Research Question, both the *full-period model*, *post-crisis model* and Holt–Winters baseline forecasts are applied and compared to actual 2026 wholesale electricity price data to determine accuracy of each model, shown in Figure 8.

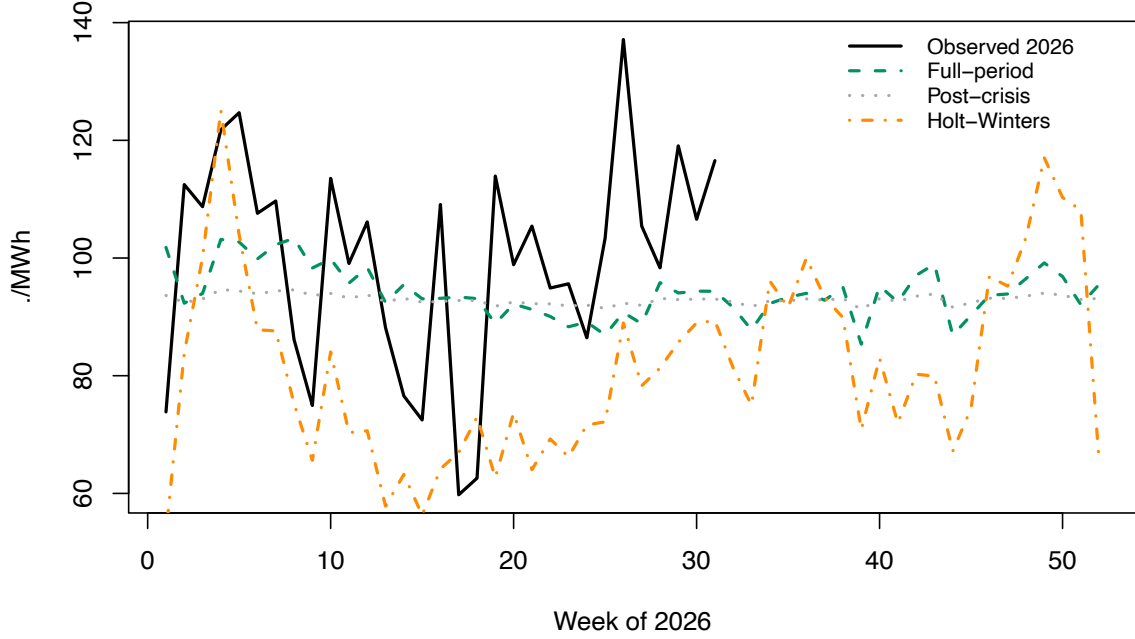


Figure 8: 2026 price forecast for *full period model*, *post-crisis model*, and Holt–Winters

The visual representation of results in Figure 8 already suggests which forecast model likely performs better. To test this numerically, I calculate the overall forecast deviation for the period from January 2026 until the first week of August 2026, a total of 31 weeks. I use the Mean Absolute Deviation (MAD) to describe the average absolute forecast error and Root Mean Squared Error (RMSE) (Hanke & Wichern 2013, pp. 36–37)⁴. The obtained values are compared against the MAD and RMSE from the Holt–Winters baseline model. These results are presented in Table 4:

⁴See appendix for MAD and RMSE formulas

Model	MAD (EUR/MWh)	RMSE (EUR/MWh)
Holt–Winters baseline	24.17	26.96
Full-period model	16.45	19.26
Post-crisis model	16.79	19.37
Relative difference	+2.1%	+0.6%

Table 4: Out-of-sample forecast accuracy for the 2026 evaluation period

The stated intuition is confirmed by the table. Both SARIMA models perform substantially better than the Holt–Winters forecast. Comparing just the SARIMA models, the post—crisis model does not predict the price developments in 2026 as accurately as the full period model. The full period model performs marginally better than the post-crisis model, with about 2.1% lower MAD and 0.6% lower RMSE. Compared to the Holt–Winters forecast, the full period model reduces MAD by approximately 32% and RMSE by 28.6%. Despite these advances in accuracy, all forecasts remain biased as their MAD values are not zero (Montgomery, Jennings & Kuhlaci 2016, p. 65). Therefore, relying on the SARIMA forecasting results, including the crisis period does not degrade forecasting accuracy of German wholesale electricity prices. Indeed, it is the opposite and marginally decreases forecasting errors.

2.8 GARCH Models

The ACF of the squared SARIMA residuals in Figure 9 indicates autocorrelation over time in the variance for both the full period and post-crisis model. As a final evaluative step, I will discuss the volatility and forecast uncertainty of both models using a Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model. This is added to complement the forecast accuracy analysis to understand the variance of forecast errors over time

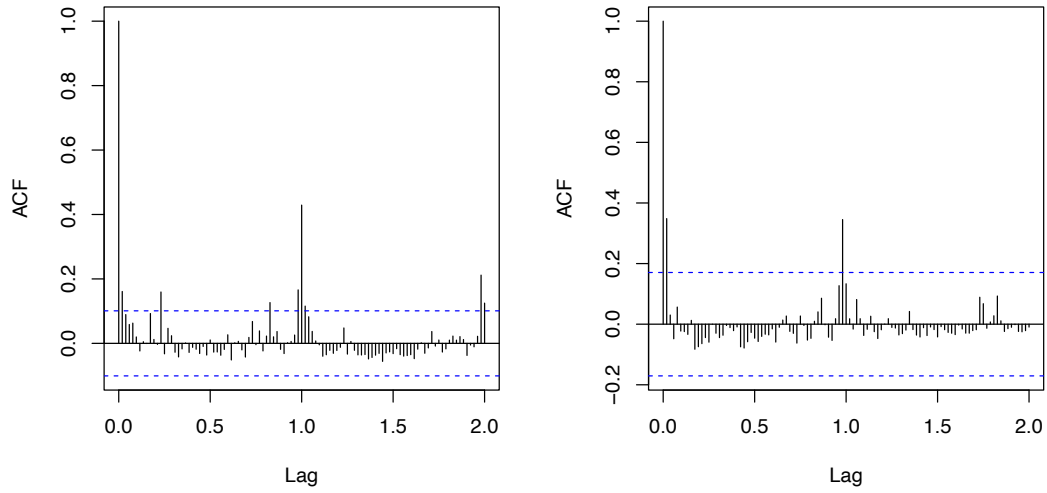


Figure 9: Correlogram of *full period* and *post-crisis* SARIMA squared residuals

	Full-period model		Post-crisis model	
	2.5%	97.5%	2.5%	97.5%
a_0	0.0130	0.0475	0.0047	0.0831
a_1	0.2284	0.4493	0.0742	0.3365
b_1	-0.0838	0.5222	-0.7803	0.7803

Table 5: 95% confidence intervals for the GARCH(1,1) parameters

Table 5 reports 95% confidence intervals for the parameters of separate GARCH(1,1) models fitted to the residuals of the full-period and post-crisis SARIMA specifications. In both models, the confidence intervals for a_1 lies completely above zero, indicating that large shocks are followed by higher volatility in the next period. For b_1 , the confidence interval does include zero, suggesting that there is no evidence of persistent volatility. Lastly, the wider confidence intervals for b_1 in the post-crisis model suggest larger estimation uncertainty in the shorter time frame.

3 Limitations & Conclusion

The results should be evaluated considering the limitations regarding methods, data, and approach. First, the post—crisis training window is not sufficiently large to derive meaningful conclusions. It includes merely 130 observations, compared to the 379 observations of the full period model. The approach is also limited by the choice of weekly observation values. The SMARD dataset also offers price data in 15—minute intervals which provides more observations and may thereby avoid problems of the current post—crisis model. It should briefly be mentioned that the given series is not suitable to interpret household energy prices. Wholesale prices and household prices differ substantially, which may limit applicability of the analysis for personal use. Aside from this, it seems that given the global developments, it might not be desirable to use a forecasting model that does deliberately excludes observations affected by price shocks and regime changes induced by geopolitical conflicts. Global geopolitical instability does not appear to fade and could influence electricity prices for years to come. The post-crisis forecasts may be better suited to predict energy prices in a “perfect world”, lacking any geopolitical tensions that may relate to global electricity price increases.

However, this rarely seems to be the case. The relatively stable energy prices from 2018 until 2022 presented a period of unusual stability that should not be naively expected for future developments in German electricity prices. Additionally, the small change in MAD and RMSE values for the full period model compared to the post-crisis model cannot be clearly attributed to either the information contained in the crisis observations or the larger number of observations. The marginal increase in forecasting accuracy over the post-crisis model can hence not be clearly attributed to the inclusion of crisis observations.

The central conclusion drawn from this project answers the research question at hand. Including crisis periods with price shocks, spikes and regime changes does not degrade forecasting accuracy of German wholesale electricity prices. Compared to the post—crisis model, the full period model produces slightly smaller forecast errors regarding the 2026 price development. The full period model and post—crisis model both exhibit substantial improvement compared to the HW forecast. Thus, the central conclusion drawn from this project answers the research question at hand. For current energy price forecasting, this has two implications. First, current weekly price forecasts may use long term data

going as far back as 2018 to improve forecasting accuracy compared to shorter data periods. Second, future price forecasting should consider models that can accommodate additional variables such as renewable energy share or geopolitical risk. Future projects should therefore consider models that can account these, such as SARIMAX models where geopolitical risk can be integrated as an additional variable (see for example Elamin & Fukushima, 2018). Other possible extensions of the project include comparisons with other countries, using higher frequency data or address remaining autocorrelation in the squared GARCH residuals. Overall, the presented analysis discusses a fruitful application of time series analysis and forecasting application regarding a widely discussed time series and offers a starting point for interesting extensions of the approach.

Appendix

MAD and RMSE formulas from Hanke & Wichern (2013, pp. 36–37):

$$\text{MAD} = \frac{1}{n} \sum_{t=1}^n \left| Y_t - \hat{Y}_t \right|$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n \left(Y_t - \hat{Y}_t \right)^2}$$

References

- Bundesnetzagentur (2026): *SMARD Marktdaten*. Available at <https://www.smard.de/home/downloadcenter/download-marktdaten/>, accessed 23 June 2026.
- Cowpertwait, P. S. P., & Metcalfe, A. V. (2009): *Introductory time series with R*. Springer. <https://doi.org/10.1007/978-0-387-88698-5>
- Eddy, M., & Alderman, L. (2022): "How Europeans Are Responding to Exorbitant Gas and Power Bills," *The New York Times*, 22 February 2022. <https://www.nytimes.com/2022/02/21/business/europe-power-gas-bill.html>, accessed 28 June 2026.
- Elamin, N. & Fukushige, M. (2018): "Modeling and forecasting hourly electricity demand by SARIMAX with interactions," *Energy*, 165, 257–268, <https://doi.org/10.1016/j.energy.2018.09.157>.
- Hanke, J. & Wichern, D. (2013): *Business Forecasting*, 9th ed. Pearson.
- Montgomery, D. C., Jennings, C. L., & Kulahci, M. (2015): *Introduction to Time Series Analysis and Forecasting*, 2nd ed. Wiley.
- Statistisches Bundesamt (2023): *Stromverbrauch der privaten Haushalte nach Haushaltsgrößenklassen*, <https://www.destatis.de/DE/Themen/Gesellschaft-Umwelt/UGR/private-haushalte/Tabellen/stromverbrauch-haushalte.html>, accessed 28 June 2026.